Kinetics Final Assignment

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February 4, 2026

Problem Two

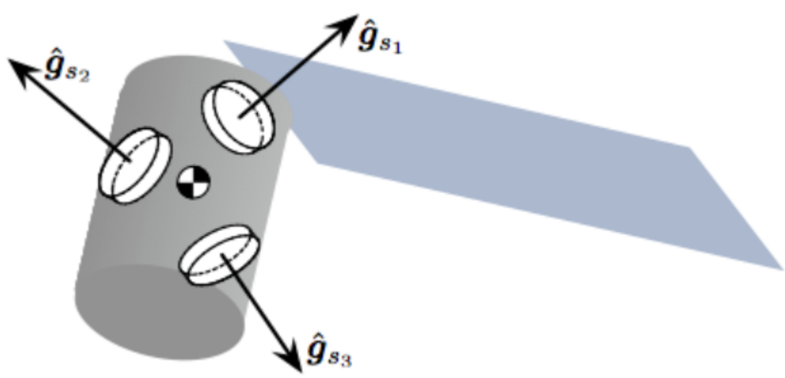

Figure 1: Illustration of Rigid Spacecraft System

Consider the rigid spacecraft with 3 Reaction Wheels (RWs) shown in Fig 1.

The wheels have generic spin axis $\hat{\mathbf{g}}_{s_i}$ which are fixed relative to the body B. Each RW is identical in build and has the same spin axis inertia I_W, while the time varying RW spin rate relative to the body is given through $\Omega_i(t)$.

Each body-fixed reaction wheel frame is defined through $\mathcal{G}_i : \{\hat{\mathbf{g}}_{s_i}, \hat{\mathbf{g}}_{t_i}, \hat{\mathbf{g}}_{g_i}\}$.

The inertia tensor $[I]$ contains the inertia of the spacecraft and the stationary wheels. Assume the spacecraft is experiencing a general disturbance $\mathbf{L}_C$ about the center of mass.

a) The angular momentum of this spacecraft about its center of mass is expressed as

$$\mathbf{H}_C = [\mathbf{I}]\omega + \mathbf{I_W} \sum_{i=1}^{3} \mathbf{\Omega_i} \hat{\mathbf{g}}_{\mathbf{s_i}}$$

Derive the rotational equations of motion for this spacecraft. (Recall that $\omega_{s_i} = \omega \cdot \hat{\mathbf{g}}_{s_i}$.)

b) Next, assume the RWs have a particular configuration where $\hat{\mathbf{g}}_{s_i} = \hat{\mathbf{b}}_i$, and $\hat{\mathbf{b}}_i$ are principal axes of the body B. Refine the general equations of motion in Part A for this scenario. You are looking for 3 scalar differential equations for ω_i.

Answer

a) We take that $\omega = {}^{\mathcal{B}}\omega_{\mathcal{B}/\mathcal{N}}$. We also know from the last problem that:

$$\dot{\mathbf{H}}_C = \mathbf{L}_C$$

So we start with taking an inertial derivative from the expression given to us:

$$\begin{aligned}\dot{\mathbf{H}}_C &= {}^{\mathcal{N}}\frac{d}{dt}\mathbf{H}_C \\ &= {}^{\mathcal{N}}\frac{d}{dt}([\mathbf{I}]\omega + \mathbf{I_W}\sum_{i=1}^{3}\mathbf{\Omega_i}\hat{\mathbf{g}}_{\mathbf{s_i}}) \\ &= {}^{\mathcal{N}}\frac{d}{dt}[\mathbf{I}]\omega + {}^{\mathcal{N}}\frac{d}{dt}\mathbf{I_W}\sum_{i=1}^{3}\mathbf{\Omega_i}\hat{\mathbf{g}}_{\mathbf{s_i}}\end{aligned}$$

The first term would result in:

$${}^{\mathcal{N}}\frac{d}{dt}[\mathbf{I}]\omega = [\mathbf{I}]\dot{\omega} + \omega \times ([\mathbf{I}]\omega)$$

Having in mind that $\hat{\mathbf{g}}_{s_i}$ are body-fixed, for the second term we will have:

$$\begin{aligned}{}^{\mathcal{N}}\frac{d}{dt}\mathbf{I_W}\sum_{i=1}^{3}\mathbf{\Omega_i}\hat{\mathbf{g}}_{\mathbf{s_i}} &= \mathbf{I_W}\sum_{i=1}^{3}{}^{\mathcal{N}}\frac{d}{dt}(\mathbf{\Omega_i}\hat{\mathbf{g}}_{\mathbf{s_i}}) \\ &= \mathbf{I_W}\sum_{i=1}^{3}(\dot{\mathbf{\Omega}}_\mathbf{i}\hat{\mathbf{g}}_{\mathbf{s_i}} + \omega \times \mathbf{\Omega_i}\hat{\mathbf{g}}_{\mathbf{s_i}})\end{aligned}$$

Putting all the parts together, the rotational equations of the motion will be in form below:

$$\begin{aligned}\dot{\mathbf{H}}_C = \mathbf{L}_C &= {}^{\mathcal{N}}\frac{d}{dt}[\mathbf{I}]\omega + {}^{\mathcal{N}}\frac{d}{dt}\mathbf{I_W}\sum_{i=1}^{3}\mathbf{\Omega_i}\hat{\mathbf{g}}_{\mathbf{s_i}} \\ &= [\mathbf{I}]\dot{\omega} + \omega \times ([\mathbf{I}]\omega) + \mathbf{I_W}\sum_{i=1}^{3}(\dot{\mathbf{\Omega}}_\mathbf{i}\hat{\mathbf{g}}_{\mathbf{s_i}} + \omega \times \mathbf{\Omega_i}\hat{\mathbf{g}}_{\mathbf{s_i}})\end{aligned}$$

b) Now, assuming that the body frame is the composed of the principal axes and they are also aligned with $\hat{\mathbf{g}}_{s_i}$, we can take the following conclusion:

$$[\mathbf{I}] = \begin{bmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{bmatrix}$$

Also let:

$$\omega = \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix}, \mathbf{\Omega} = \begin{bmatrix} \Omega_1 \\ \Omega_2 \\ \Omega_3 \end{bmatrix}$$

After doing the calculations for each $\hat{\mathbf{g}}_{s_i}$, we'll end up with the three scalar differential equations for ω_i:

$$i = 1 : I_1\dot{\omega}_1 = L_1 - (I_3 - I_2)\omega_3\omega_2 - I_W\dot{\Omega}_1 - I_W(\omega_2\Omega_3 - \omega_3\Omega_2)$$

$$i = 2 : I_2\dot{\omega}_2 = L_2 - (I_1 - I_3)\omega_1\omega_3 - I_W\dot{\Omega}_2 - I_W(\omega_3\Omega_1 - \omega_1\Omega_3)$$

$$i = 3 : I_3\dot{\omega}_3 = L_3 - (I_2 - I_1)\omega_1\omega_2 - I_W\dot{\Omega}_3 - I_W(\omega_1\Omega_2 - \omega_2\Omega_1)$$